PHY301 Assignment#2 Solution 2023

by Artist H 25

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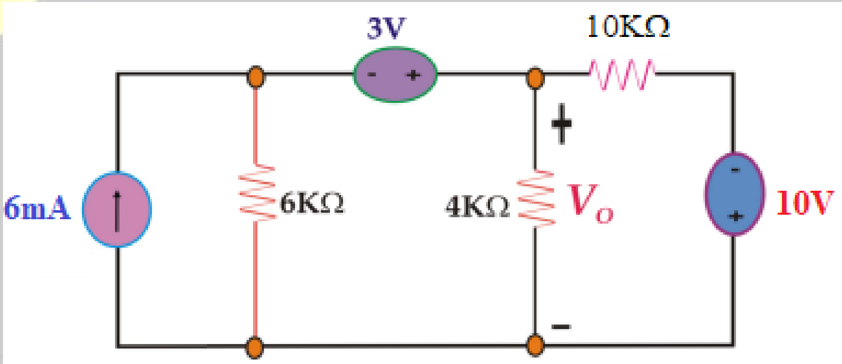
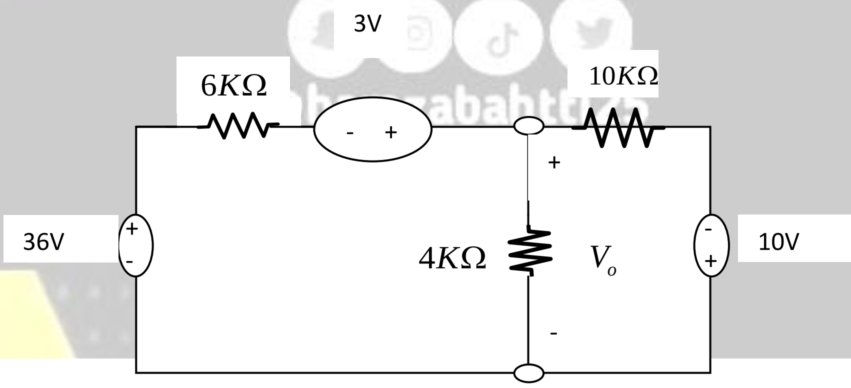
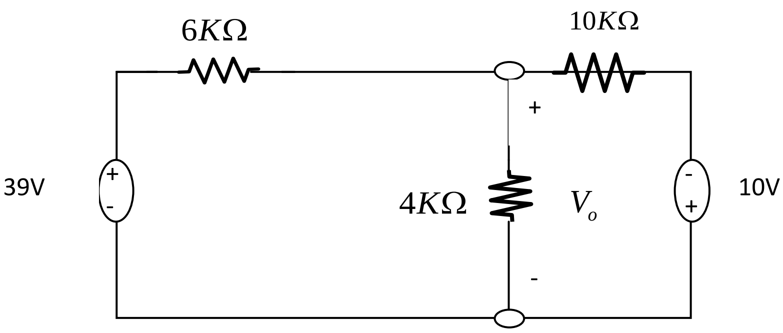
Calculate V_O by source transformation & Loop method. Draw and label the circuit diagram of each step, also mention the unit of each derived value. [Marks:10]



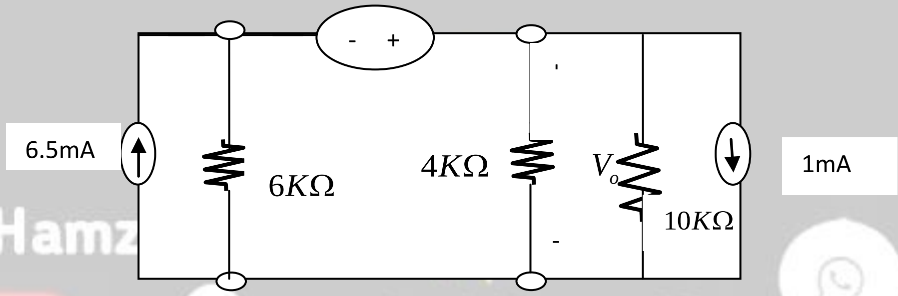
Solution:



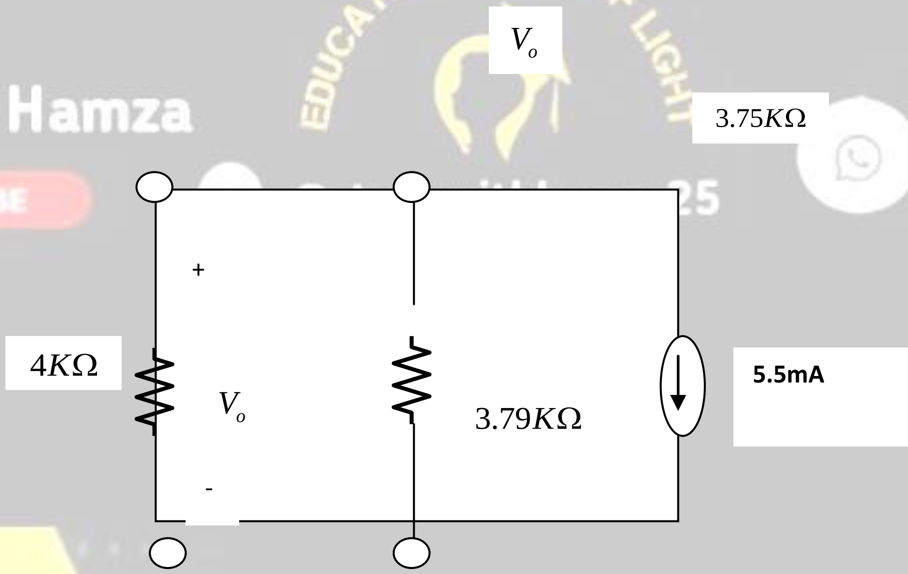
V=IR imply that v=6×6=36V



I=V /t =39/6=6.5mA and   10/10=1mA



Rt=R1×R2/R1+R2 = 6×10/6+10=3.75



Apply current division Rule:

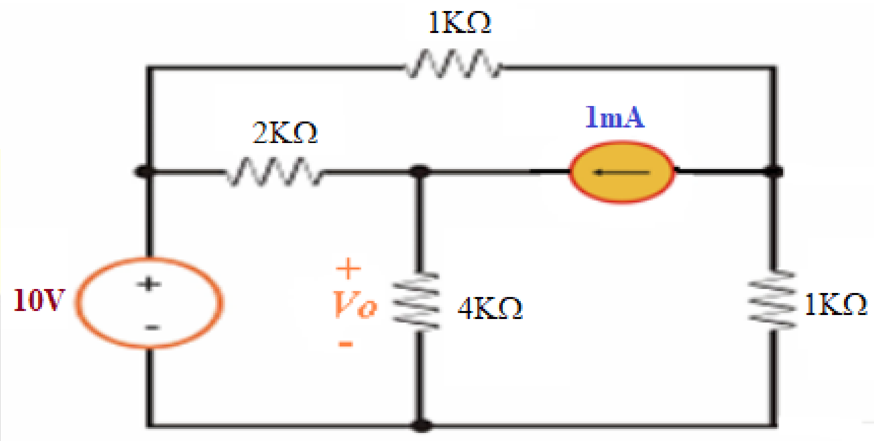
$$I_{4K\Omega} = 5.5mA \times \frac{3.75}{3.75+4} K\Omega = 2.66mA$$

$$V_o = 2.66 \times 4$$

$$= 10.64v$$

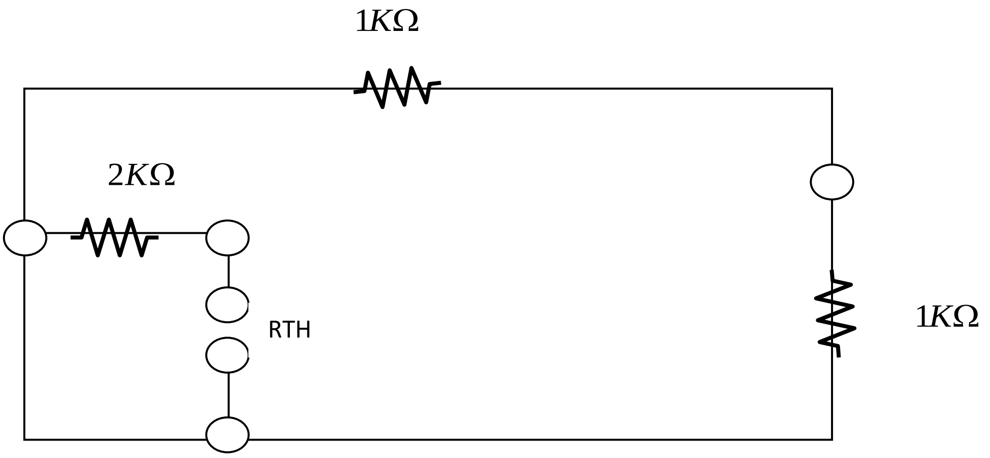
Q.2:

Use Thevenin’s theorem to find *Vo* in the network given below.
(Write each step of the calculation & draw circuit diagram where required to get maximum marks)



/10KΩ

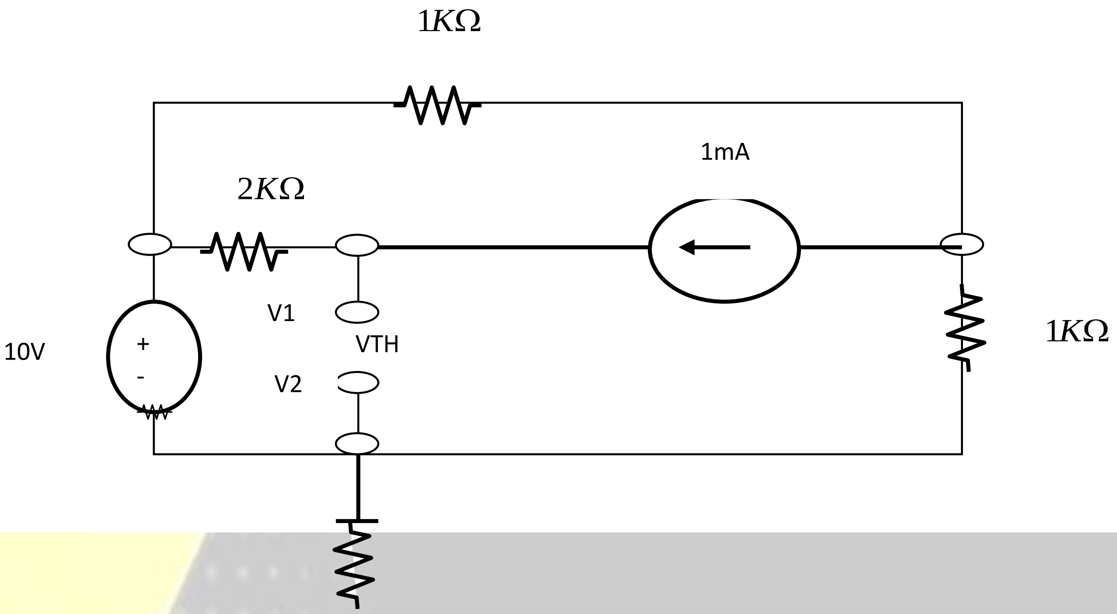
Solution:



$$R_{TH} = 2K\Omega$$

Because in terminal a and b circuit is open in series resistance will be =0

To Find VTH:



KVL to super mesh

$$-10\times 1\times 10^{3}\times I_1+1\times 10^{3}\times I_2=0$$
$$10^{3}\times I_1\times 10^{3}\times I_2=10$$

Equation 1:

$$10^{3}\times I_1\times 10^{3}\times I_2=10$$

From Super Mesh:

$$I_1-I_2=1\times 10^{-3}$$

Equation 2:

$$I_1-I_2=1\times 10^{-3}$$

Solving Equation 1 & 2

$$I_1 = 5.5mA$$

$$I_2 = 4.5mA$$

Apply KVL to MESH 3

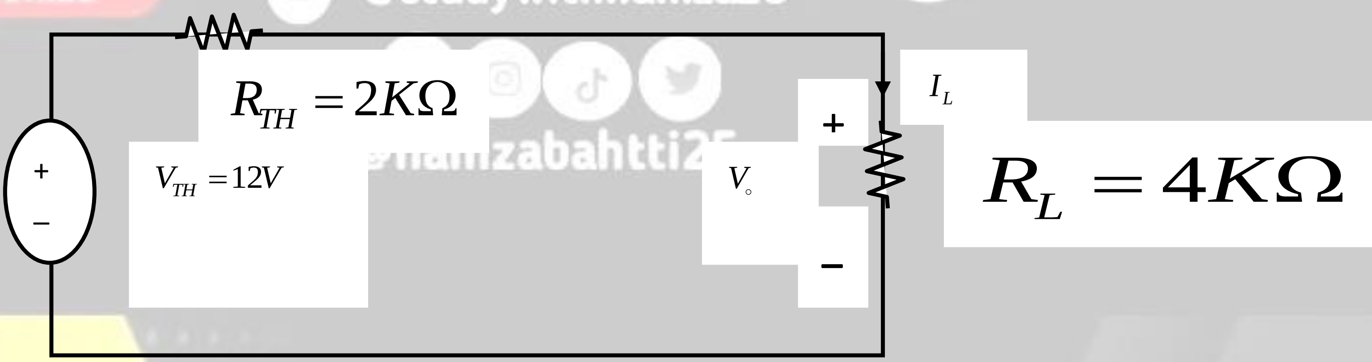
$$-10 + 2\times10^3(I_2 - I_1) + V_{\circ c} = 0$$

$$V_{\circ c} = 10 + 2\times10^3(4.5\times10^{-3} - 5.5\times10^{-3})$$

$$V_{\circ c} = 12V$$

$$V_{TH} = 12V$$

Thevenin's theorem to find *Vo*



The venin's theorem Circuit

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

$$= \frac{12}{2\times10^3 + 4\times10^3}$$

$$I_L = 2\times10^{-3}\, mA$$

$$V_o = I_L \times R_L$$

$$= 2\times10^3 \times 4\times10^3$$

$$V_o = 8V$$